# AI Research Collaboration System using Multi-Agent Systems

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## 1. Introduction

Collaboration between researchers is essential for innovation, especially in rapidly evolving fields such as artificial intelligence. However, identifying suitable collaborators is difficult due to the large volume of research output and the diversity of expertise.

This project introduces a Multi-Agent System that automates the process of identifying collaboration opportunities. The system analyzes research data, builds structured profiles, computes similarity between researchers, and generates collaboration recommendations.

An interactive dashboard was developed to visualize and explore the results.

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## 2. System Architecture

The system is designed as a pipeline of intelligent agents, each responsible for a specific task.

### Architecture Overview

Observer → Profiler → Matcher → Negotiator → Dashboard

- Observer collects research data
- Profiler builds researcher profiles
- Matcher computes similarity scores
- Negotiator makes collaboration decisions
- Dashboard visualizes results

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## 3. Agent Descriptions

### AgentObserver

Collects research data including papers and authors. It acts as the entry point of the system.

### AgentProfiler

Builds structured researcher profiles using:

- publications count
- citations
- expertise

### AgentMatcher

Computes similarity between researchers and assigns a collaboration score based on compatibility.

### AgentNegotiator

Determines whether a collaboration is accepted or rejected using scoring rules.

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## 4. Workflow

1. Collect research papers
2. Build researcher profiles
3. Compute similarity scores
4. Generate collaboration candidates
5. Apply negotiation rules
6. Store results
7. Visualize through dashboard

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## 5. Data and Database

The system produces a dataset stored as a CSV file.

Main fields include:

- researcher 1
- researcher 2
- collaboration score
- decision

This dataset acts as a lightweight database for analysis and visualization.

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# 6. Results

The system successfully identifies high quality collaboration opportunities.

Key results:

- ranked collaboration recommendations
- accepted collaborations
- similarity scores

Example:

- Dzmitry Bahdanau and Yoshua Bengio → high score
- Kyunghyun Cho and Yoshua Bengio → strong match

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# 7. Web Interface and Dashboard

The system includes an interactive dashboard built using Streamlit.

### Features:

- total collaborations metric
- accepted collaborations metric
- average score metric
- collaboration table
- filtering options
- score visualization
- network graph

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**Figure 1: AI Research Collaboration Dashboard**

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# 8. Conclusion

This project demonstrates how Multi-Agent Systems can be used to automate collaboration discovery.

The system integrates data processing, profiling, similarity analysis, and decision making into a unified pipeline.

It provides a scalable approach to analyzing research ecosystems.

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# 9. Future Work

- integrate real APIs such as arXiv or Semantic Scholar
- improve similarity using machine learning
- deploy as a cloud application
- enable real time recommendations

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# 10. References


- Streamlit documentation

- NetworkX documentation
- Python libraries